AI For Bharat

AquaAlert

- **Problem Statement Title:–** Smart Community Health Monitoring and Early Warning System for Water-Borne Diseases in Rural Northeast India
- **Theme:–** HealthCare
- **PS Category:–** Software
- **Team Name:–** BlackShadow
- **Team Members:–**
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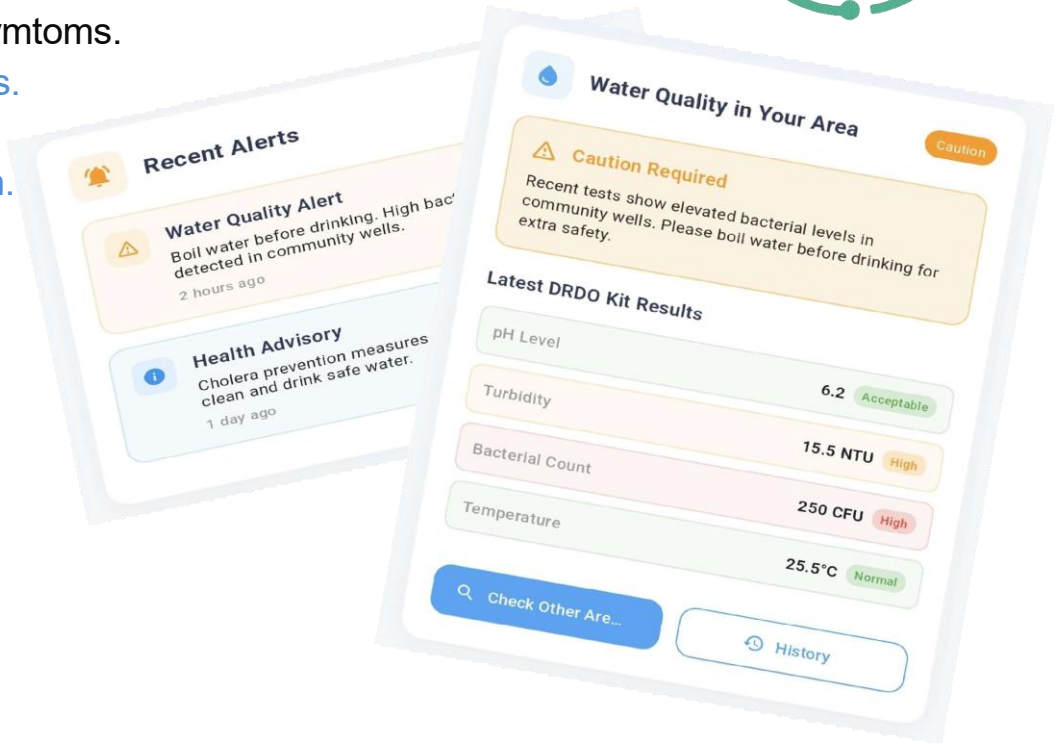
AquaAlert

WHY THIS APP ?

- To spread knowledge on how this disease spread and their symtoms.
- To prevent outbreak by giving timely alert about affected areas.
- To promote safe water practices and hygiene habits.
- To track areas where water-borne diseases are most common.
- To reduce health risk through timely and reliable information.



PROBLEMS FACED WITHOUT THE APP:

- Lack of awareness
- Delayed treatment
- No early warning system
- High spread of diseases
- Poor community preparedness

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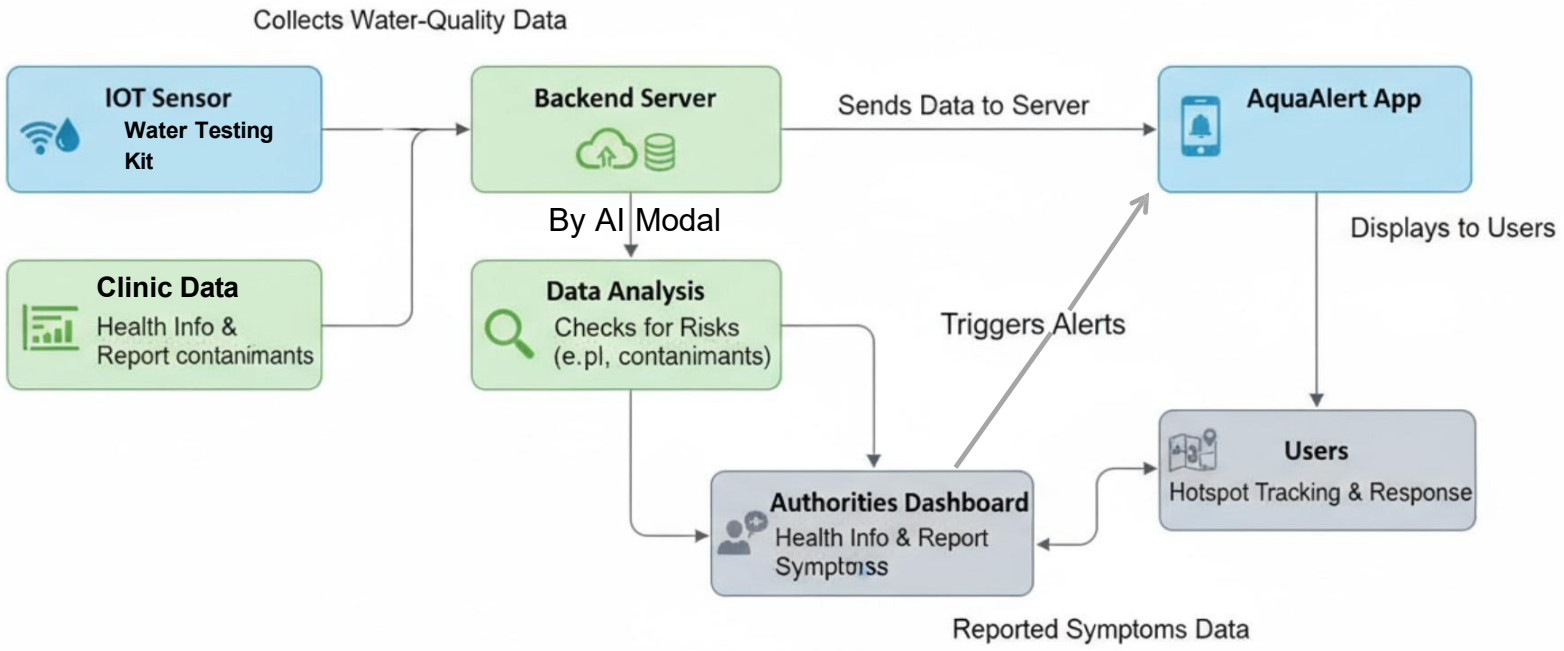
TECHNICAL APPROACH

- Technology to be used (e.g. Dart, Flutter, IoT Sensors, Serverpod, Cloud Server, Python(AI Devlopment), Database: MongoDB, Twilio API (WhatsApp))



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FEASIBILITY AND VIABILITY

Feasablity of Idea:

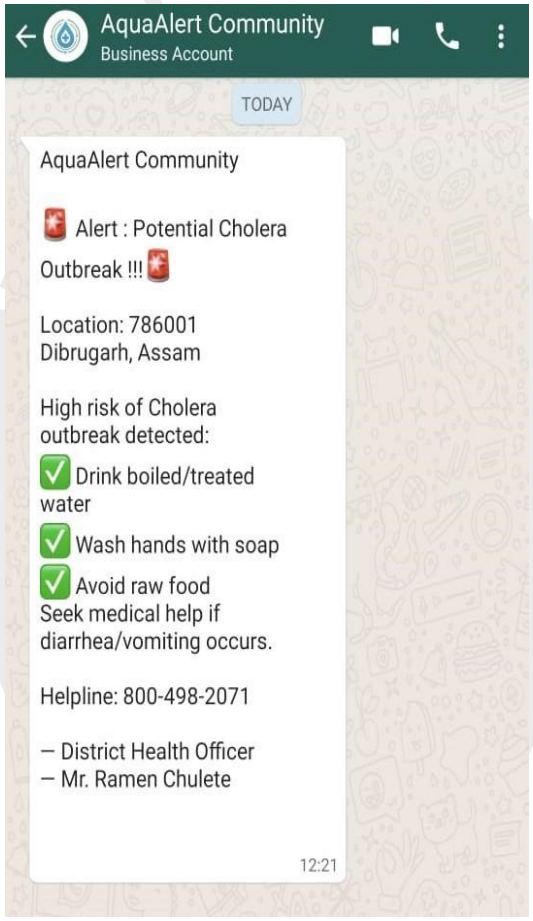
- Technological Readiness
- User Acceptance
- Cost-effectiveness
- Microservices

Potential Challenges and Risks:

- Data Quality and Reliability
- Connectivity Issues
- Resource Constraints
- Community Engagement

Strategies for Overcoming the Challenges:

- Robust Sensor Design & Training
- Offline-first Design
- Energy OptimizationLocalization & Awareness



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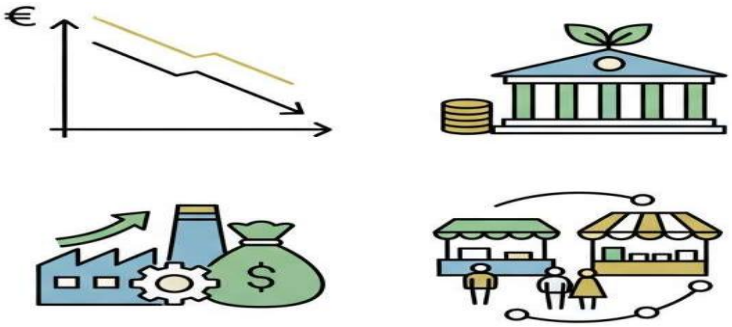
IMPACT AND BENEFITS

SOCIAL:

- Awareness and Education
- Community Safety
- Improved Health Outcomes
- Trust in Technology

ECONOMIC:

- Reduce Healthcare Costs
- Saves Government Resources
- Productivity Gains
- Support for Local Economy



ENVIRONMENTAL:

- Better Water Management
- Promotes Clean Practises
- Sustainable Habits
- Indirect Pollution Control

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RESEARCH AND REFERENCES

For Data Collection for making old seasonal trends:

➢ **Primary Health Surveillance Databases**

- **Integrated Disease Surveillance Programme (IDSP): https://idsp.mohfw.gov.in**
- **Health Management Information System (HMIS): https://hmis.mohfw.gov.in**
- **Water Quality Monitoring Information System (WQMIS): https://abdm.gov.in**
- **Central Pollution Control Board (CPCB) Water Quality Database: https://www.cseindia.org**
- **Water Quality and Environmental Monitoring(Jal Jeevan Mission): https://jaljeevanmission.gov.in**

Tribal and Regional Health Databases

- **Swasthya Tribal Health Portal: https://swasthya.tribal.gov.in**
- **Northeast India Tribal Population Database: https://nhm.gov.in/index1.php?lang=1&level=1&sublinkid=150&lid=226**

Community Health Worker Systems

- **ASHA Worker Data Collection System: https://nhm.gov.in/index1.php?lang=1&level=1&sublinkid=150&lid=226**
- **National Health Mission (NHM) ASHA Database: https://www.data.gov.in/keywords/asha**

Disease-Specific Databases

- **Open Government Data Platform - Water-borne Diseases: https://www.data.gov.in/keywords/Water-borne%20diseases**
- **National Centre for Vector Borne Diseases Control (NCVBDC): https://ncvbdc.mohfw.gov.in**
- **ICMR Food and Water-borne Disease Surveillance Database: https://www.icmr.gov.in**